
JevCal: A Benchmark for Calibration and Semantic Evaluation of Jev-like Models

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Abstract

Jev-like decision models expose typed probabilistic interfaces: given a state and a bounded question, they return a probability distribution rather than free-form text. Existing JevBench-style evaluations measure decision accuracy, schema validity, latency, cost, Brier score, and top-label expected calibration error, but their calibration view is incomplete for this setting because it evaluates only a projection of the returned distribution. We introduce JevCal, a benchmark for pointwise probability fidelity, representation sensitivity, and cross-primitive semantic consistency. JevCal procedurally generates probability problems with known ground truth, renders each latent problem in multiple equivalent forms, and asks the same binary proposition through three Jev primitives: Noul, Choice, and Score. In a development run on 365 rendered items with 10 repeated calls per primitive, Noul is substantially closer to the exact event probability than binary Choice: mean total variation error is 0.0817 for Noul, 0.1139 for Score expectation, and 0.2362 for Choice. Choice is systematically more True-heavy, with mean Choice-minus-Noul probability 0.179. Latent-level one-sided tests reject zero implied unknown mass and zero Score-unknown improvement. An open SemIf/Qwen3.5-4B binary baseline scores 0.371 mean TV, illustrating that JevCal is also useful for non-Jev typed-decision systems.

1 INTRODUCTION

Many deployed AI systems need small, repeated decisions rather than open-ended generation. Jev-like models target this regime by accepting structured state and a bounded typed question, then returning a distribution over admissible answers. This interface is attractive because downstream agents can make thresholded decisions, route uncertainty to a human, and log probabilistic evidence without parsing prose.

The central question, however, is not only whether the argmax answer is correct. If an interface returns probabilities, users need to know what those probabilities mean. A model can have high argmax accuracy while assigning probabilities that are not recoverable from the state, not invariant to representation, or not semantically comparable across typed primitives.

We introduce *JevCal*, a benchmark for Jev-like probability semantics. JevCal evaluates four separable properties:

1. **Pointwise probability fidelity:** when the state fully specifies an exact conditional probability $p^*(x)$, does the model return $\hat{p}(x) \approx p^*(x)$?
2. **Representation sensitivity:** does $\hat{p}$ change when the same latent problem is rendered as counts, ratios, tables, prose, or nested state?
3. **Primitive semantics:** do different typed primitives expose the same probabilistic object?
4. **Empirical calibration diagnostics:** when labels are sampled or observed, do forecast probabilities match frequencies under the chosen projection?

Our main empirical finding is that, in the tested Jev endpoint, Noul and Score appear much closer to exact event probability than binary Choice. Choice often behaves as if it renormalizes after removing an implicit middle or unknown mass. We treat this as a black-box behavioral description, not as an architectural claim.

2 BACKGROUND AND RELATED WORK

2.1 JevBench

JevBench is a public benchmark for Jev-class typed decision models. Its repository describes a suite in which models see the same state, instructions, rubric, and exact label set; systems are scored on accuracy, schema validity, Brier score, expected calibration error (ECE), ordinal MAE, paraphrase consistency, latency, cost, and openness [JevBench Repository(2026)]. The public v1.0 description contains 242 decisions across request routing, answer adequacy judging, policy checks, intent classification, ordinal severity scoring, and enum extraction, with a mixture of public, held-out, and imported-source decisions [JevBench Repository(2026)]. The same repository states that ECE is computed as top-label confidence with 10 equal-width bins [JevBench Repository(2026)]. A separate JevBench site describes an interactive-agent harness roadmap and notes that other Jev-focused text-decision benchmarks exist elsewhere [JevBench Site(2026)].

This makes JevBench valuable as a broad decision-model leaderboard, but its calibration definition is not complete for probabilistic typed interfaces. Top-label ECE asks whether, among examples where a particular selected label has confidence q , that selected label is correct about q of the time. It does not test whether all non-argmax probabilities are meaningful, whether equivalent states produce the same distribution, or whether two typed primitives assign compatible probabilities to the same proposition.

At the time of writing, we found public JevBench repositories, websites, and informal audits, but no formal arXiv-style JevBench preprint dedicated to the calibration semantics studied here.

2.2 Five Notions of Calibration

Classical probabilistic calibration asks whether events assigned probability p occur with frequency p [Dawid(1982)]. For a binary event, this is the scalar condition

$$P(Y = 1 \mid \hat{p} = p) = p.$$

Proper scores such as the Brier and logarithmic scores reward truthful distributions in expectation [Brier(1950), Gneiting and Raftery(2007)]. However, for multiclass classifiers the word calibration refers to several genuinely different projections [Vaicenavicius et al.(2019), Gupta and Ramdas(2021)].

Let a model output $\hat{\mathbf{P}}(X) = (\hat{P}_1, \dots, \hat{P}_K)$. *Full multi-*

class calibration requires

$$P(Y = k \mid \hat{\mathbf{P}} = \mathbf{p}) = p_k$$

for each class k . *Classwise calibration* weakens the conditioning to one coordinate:

$$P(Y = k \mid \hat{P}_k = p) = p.$$

This is the relevant question when users may act on any individual Choice probability rather than only on the winning option; Kull et al. explicitly distinguish confidence-ECE and classwise-ECE in multiclass calibration experiments [Kull et al.(2019)].

Confidence calibration, popularized in modern neural-network evaluation, defines

$$\hat{Y} = \arg \max_k \hat{P}_k, \quad C = \max_k \hat{P}_k$$

and asks whether

$$P(Y = \hat{Y} \mid C = c) = c.$$

Standard ECE is usually an estimator of this quantity [Guo et al.(2017)]. *Top-label calibration* conditions also on the predicted class:

$$P(Y = k \mid \hat{Y} = k, C = c) = c.$$

This is stronger than confidence calibration but still inspects only the maximum-probability coordinate [Gupta and Ramdas(2021)]. Nixon et al. show that conclusions can change with the calibration projection, binning rule, class conditioning, and norm [Nixon et al.(2019)]; Kumar et al. further show that common calibration-error estimators can have poor sample properties [Kumar et al.(2019)].

Binary Noul is special: one coordinate determines the other, so classwise, full binary, and scalar probability calibration nearly collapse. Choice is different in the general case because it returns a categorical distribution. Even for binary Choice, JevCal can ask a question ordinary calibration cannot: since the generator gives the exact conditional probability $p^*(x)$, we can measure pointwise probability fidelity $\hat{p}(x) \approx p^*(x)$ rather than only empirical calibration against one sampled label.

3 THE JEVCAL BENCHMARK

3.1 Dataset Construction

Each JevCal item begins as a latent probabilistic problem with exact ground truth. The current dataset uses six procedural families:

- explicit probability, where p^* is directly supplied;

- frequency recovery from counts;
- compound probability, including complements, intersections, unions, repeated trials, and sampling without replacement;
- conditional probability over contingency tables and filtered populations;
- Bayes problems with exact posteriors;
- sequential Bayes problems with multiple updates.

Each latent item is rendered in one or more equivalent representations: direct probabilities, counts, scaled counts, ratios, tables, prose, nested state, and distractor-augmented state. All renderings of the same latent item share the same gold distribution. This design lets us separate mathematical recovery failures from representation sensitivity.

The fixed release used for the parallel primitive results is `v0.1.0_parallel_primitives_dev.jsonl`, with 365 rows and SHA-256 prefix `2f31c498`; the full hash is recorded in the release manifest. It contains 92 latent problems and 365 rendered items. Family counts are: 80 explicit-probability, 30 frequency, 60 compound, 100 conditional, 35 Bayes, and 60 sequential-Bayes rendered items. The ordinary probability-recovery release `v0.1.0_dev.jsonl` contains 2,600 rendered exact-probability items.

3.2 Parallel Primitive Contract

For every rendered binary item, JevCal stores one shared state, one shared proposition A , and the exact binary gold probability

$$p^* = P(A), \quad 1 - p^* = P(\neg A).$$

The same state and proposition are then queried through three typed primitives:

- **Noul**: returns a probability that the proposition is true;
- **Choice**: returns a distribution over `False` and `True`;
- **Score**: returns a distribution over 10 ordered truth levels with normalized values $z_j = j/9$.

The Score item does not encode a gold distribution over levels. Instead, evaluation uses the expected truth value

$$\mu_S = \sum_{j=0}^9 q_j z_j$$

as one black-box projection from Score to event probability.

Table 1: Probability recovery by primitive. TV equals binary MAE. Results use per-item mean predictions over 10 repeats.

Primitive	n	Mean TV	Median TV	Accuracy
Choice	365	0.236	0.229	0.693
Noul	365	0.0817	0.0508	0.893
Score exp.	365	0.1139	0.0813	0.893

Table 2: Cross-primitive semantic distances.

Comparison	Mean abs. diff.	Median
$ p_N - p_C $	0.192	0.189
$ p_N - \mu_S $	0.0541	0.0384
$ p_C - \mu_S $	0.207	0.210
$p_C - p_N$	0.179	0.189

3.3 Metrics

For binary probability recovery, total variation distance is equal to absolute error:

$$\text{TV}(\hat{p}, p^*) = |\hat{p} - p^*|.$$

We also report Brier regret, KL/log-score regret with clipping, argmax accuracy as a secondary decision metric, and representation sensitivity. Repeated calls are preserved during collection and averaged only for headline recovery tables. In the Jev development run, each of the 365 rendered items was queried 10 times for each primitive, yielding 10,950 raw result rows.

To study cross-primitive semantics, we compare p_N , p_C , and μ_S , where p_N is Noul’s probability of truth, p_C is Choice’s probability of `True`, and μ_S is Score expectation. We also fit black-box mappings from Noul to Choice and from Score to Choice on held-out analyses.

4 RESULTS

Table 1 shows the core result. Noul has the lowest mean probability-recovery error, followed by Score expectation. Binary Choice has almost three times the mean TV error of Noul. The gap is not only an argmax issue: Choice has lower decision accuracy, but its probability error remains the more diagnostic failure.

Table 2 shows that Score expectation is much closer to Noul than to Choice. Choice is systematically True-heavy relative to Noul: the mean signed difference $p_C - p_N$ is 0.179.

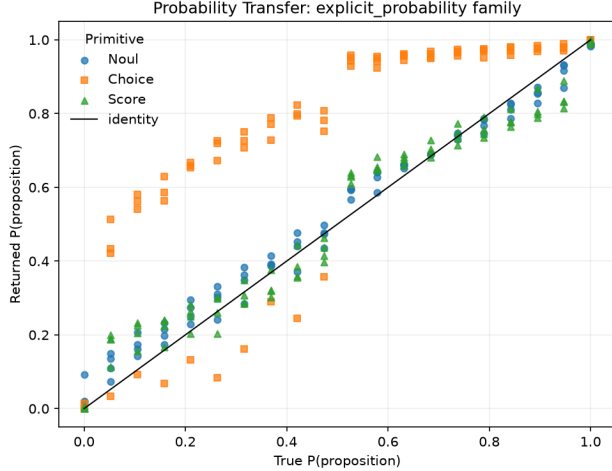


Figure 1: Probability transfer on explicit-probability items. Noul and Score expectation remain near the identity curve; Choice saturates toward high True probabilities across much of the middle range.

4.1 Representation Sensitivity

JevCal also exposes representation effects. Mean pairwise TV across equivalent renderings is 0.044 for Noul, 0.0656 for Score expectation, and 0.0998 for Choice. Thus the primitive with the weakest probability recovery is also the most representation-sensitive in this run.

The largest recovery errors occur on compound problems. Noul mean TV is 0.036 on explicit-probability items and 0.030 on frequency items, but 0.205 on compound items. Choice is worse across all families, peaking at 0.323 mean TV on compound items.

4.2 Open SemIf Baseline

We also evaluated the SemIf Qwen3.5-4B direct-logit system on the 365 binary items, using one response per item. This is not a Noul/Choice/Score triple, so it does not support the cross-primitive analyses above. It is nevertheless informative as an open Jev-like typed-decision baseline: mean TV is 0.371, median TV is 0.335, argmax accuracy is 0.485, empirical ECE is 0.354, and mean representation sensitivity is 0.117 pairwise TV. The strongest family is conditional probability (mean TV 0.257), while compound probability is weakest (mean TV 0.515). These results reinforce the benchmark’s intended use: typed option probabilities from an open model should not be assumed calibrated or semantically faithful without workload-specific validation.

4.3 Noul to Choice Calibration

If Choice and Noul exposed the same event probability, $p_C \approx p_N$. This identity mapping performs poorly on the held-out mapping analysis: MAE is 0.194 and R^2 is 0.169. A simple power mapping

$$p_C \approx 1 - (1 - p_N)^\alpha$$

with $\alpha = 2$ reduces MAE to 0.074 and yields $R^2 = 0.800$. An isotonic mapping is slightly better, with MAE 0.0705 and $R^2 = 0.817$.

This suggests that Choice is not merely noisy around Noul. It follows a systematic distortion that pushes middle probabilities toward True.

4.4 Score, Choice, and an Implicit Unknown Mass

Score provides a useful behavioral explanation. Suppose a Score distribution is decomposed into three masses:

$$T + U + F = 1,$$

where T is true-like mass, F is false-like mass, and U is an implied middle or unknown mass. Then binary Choice may behave approximately as

$$p_C \approx \frac{T}{T + F} = \frac{T}{1 - U}.$$

Using Score expectation alone to predict Choice gives MAE 0.221. Adding a central-decay unknown mass reduces MAE to 0.0955. A learned symmetric unknown profile reduces MAE further to 0.0733 with $R^2 = 0.821$ and correlation 0.920. This is comparable to the best Noul-to-Choice mappings.

We also tested the null hypothesis that the unknown-mass effect is zero. To reduce dependence from multiple renderings of the same latent problem, we report latent-mean tests as the primary statistics. The Noul/Choice-implied unknown mass is positive over latent means: mean 0.250, 95% CI [0.223, 0.277], $t(91) = 18.08$, one-sided $p = 3.5 \times 10^{-32}$. The held-out Score-derived unknown mass is also positive: mean 0.291, 95% CI [0.255, 0.328], $t(21) = 15.63$, $p = 2.4 \times 10^{-13}$. Finally, adding Score-derived unknown mass improves absolute Choice prediction error: mean improvement 0.118, 95% CI [0.0847, 0.152], $t(21) = 6.90$, $p = 4.0 \times 10^{-7}$. These tests reject the null of no unknown mass, and also reject the weaker null that the Score-derived unknown component provides no predictive improvement.

The interpretation is intentionally operational. We do not claim that Jev internally stores a three-valued truth representation. We claim that observed Noul,

Score Expectation Compared with Gold, Noul, and Choice

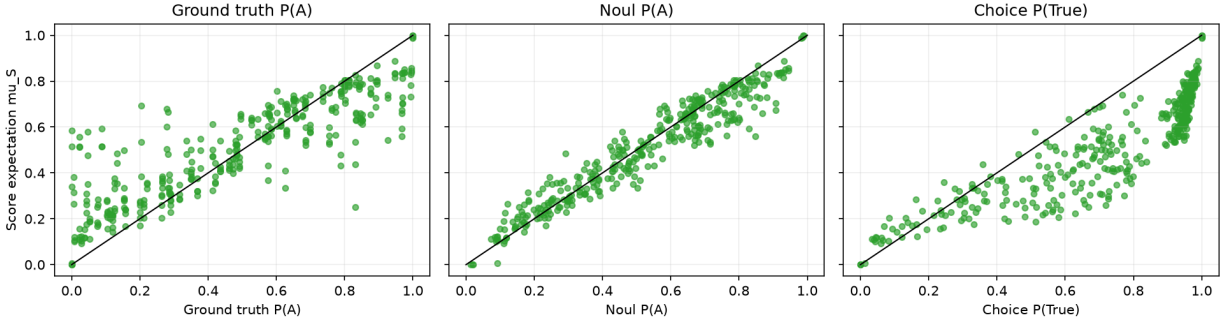


Figure 2: Score expectation compared with ground truth, Noul, and Choice. The middle panel shows the closest alignment: Score expectation tracks Noul more tightly than it tracks binary Choice.

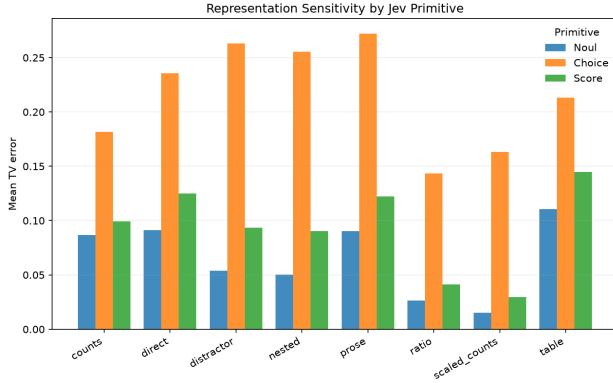


Figure 3: Representation sensitivity by primitive. Choice is consistently more sensitive to equivalent renderings than Noul or Score expectation.

Choice, and Score behavior is consistent with Choice acting like a forced binary projection of a richer truth-status distribution.

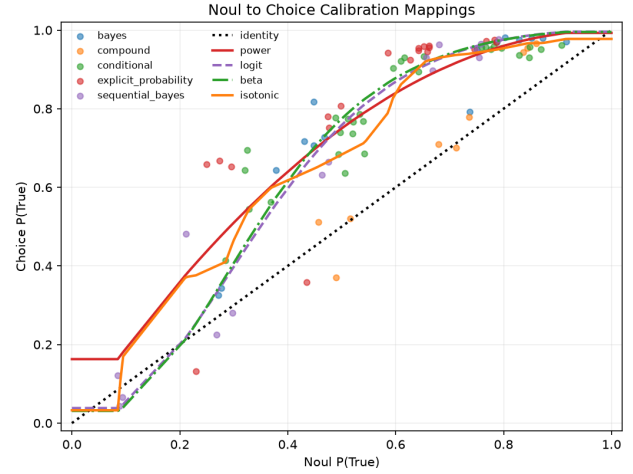


Figure 4: Noul-to-Choice mappings. The identity line is a poor explanation of Choice; a simple power mapping and isotonic fit capture much of the distortion.

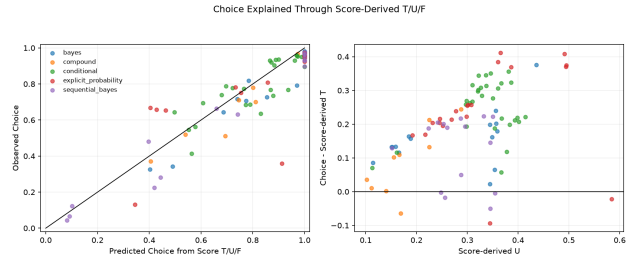


Figure 5: Predicting Choice from Score-derived truth/unknown/false mass. The left panel compares predicted and observed Choice probability; the right panel shows residuals against derived unknown mass.

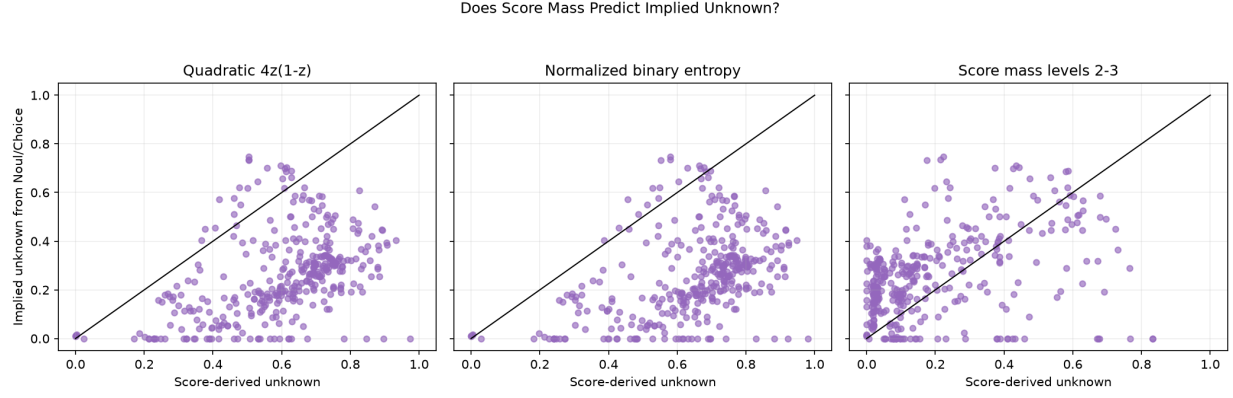


Figure 6: Several Score-derived unknown-mass proxies compared with the implied unknown mass from Noul/Choice. Middle-level Score mass and entropy-like constructions carry signal about the gap between Noul and Choice.

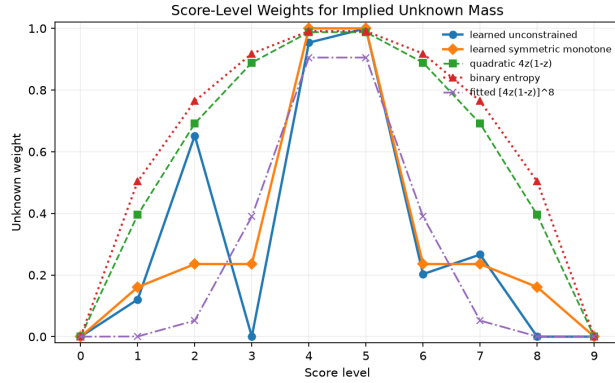


Figure 7: Score-level weights for predicting implied unknown mass. Learned and structured profiles place most unknown weight near the middle truth levels, rather than at the endpoints.

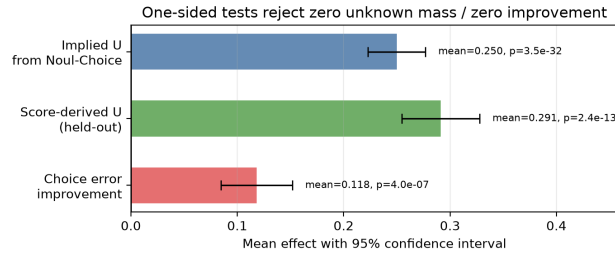


Figure 8: Latent-level one-sided tests for unknown-mass effects. Error bars show 95% confidence intervals. All displayed intervals are bounded away from zero.

5 DISCUSSION

JevCal complements JevBench-style leaderboards rather than replacing them. JevBench asks whether systems are useful typed decision models under realistic constraints such as validity, speed, and cost. JevCal asks a narrower semantic question: when the right probability is exactly known, what does the returned distribution mean?

This distinction matters because confidence or top-label calibration can be satisfied while semantically important probability mass is wrong. A binary Choice interface may look calibrated under an argmax-confidence view while still failing to preserve the event probability supplied by an equivalent Noul query. Similarly, a model may have acceptable average Brier score but change its probabilities under representation-preserving rewrites.

For agentic use, these failures have different consequences. Argmax accuracy matters when only the chosen action is used. Probability recovery matters when thresholds, abstention policies, expected utilities, or downstream Bayesian updates depend on the returned number. Cross-primitive consistency matters when an API offers multiple typed ways to ask what appears to be the same probabilistic question.

6 LIMITATIONS

This paper reports a development run, not a final leaderboard. The dataset has 365 rendered items from 92 latent problems, all synthetic and binary after conversion to the parallel primitive contract. Larger releases should add more latent seeds, multiclass exact-probability problems, additional Jev-like systems, and held-out public/private splits.

The fitted Noul-to-Choice and Score-to-Choice mappings are descriptive. They establish systematic black-box relationships in one run but should not be read as causal or architectural explanations. The current Score analysis also depends on a chosen 10-level rubric and on projection rules that may not transfer to other score schemas.

Finally, pointwise probability fidelity is not a substitute for empirical outcome calibration. JevCal states the true probability in or through the input state and asks whether the model recovers it. Classical calibration asks whether realized stochastic events occur at the predicted frequency, and its empirical estimates depend on sample size and binning. A complete evaluation of Jev-like models should include both.

7 CONCLUSION

JevCal is a benchmark for evaluating the semantics of probabilistic typed decision models. It procedurally generates exact probability problems, renders them in equivalent forms, and compares Noul, Choice, and Score on the same proposition. In our development run, Noul best recovers the exact event probability, Score expectation closely follows Noul, and binary Choice is systematically True-heavy. A Score-derived unknown-mass model explains much of the Choice distortion. These findings motivate pointwise probability fidelity, representation sensitivity, empirical calibration diagnostics, and cross-primitive semantic consistency as first-class metrics for Jev-like systems.

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